

Soldering of Iron Catalysts for Direct Synthesis of Light Olefins from Syngas under Mild Reaction Conditions

Vitaly V. Ordonsky,^{†,||} Yuan Luo,[†] Bang Gu,[†] Alexandre Carvalho,[†] Petr A. Chernavskii,[‡] Kang Cheng,[§] and Andrei Y. Khodakov^{*,†,||}

[†]Unité de Catalyse et Chimie du Solide, UMR 8181 CNRS, Bât. C3, Université Lille 1, ENSCL, Ecole Centrale de Lille, 59655 Villeneuve d'Ascq, France

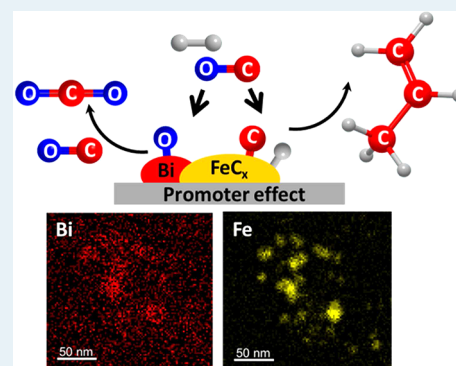
[‡]Department of Chemistry, Moscow State University, 119992 Moscow, Russia

[§]Collaborative Innovation Center of Chemistry for Energy Materials, State Key Laboratory of Physical Chemistry of Solid Surfaces, Xiamen University, 361000 Xiamen, People's Republic of China

S Supporting Information

ABSTRACT: High-temperature Fischer–Tropsch synthesis represents a sustainable alternative for direct light olefin synthesis from syngas derived from fossil and renewable feedstocks. It is found that the promotion of iron catalysts with metals used for soldering (Bi and Pb) results in a remarkable increase in the light olefin production rate with a possibility to conduct Fischer–Tropsch synthesis at low reaction pressure. A combination of characterization techniques uncovered notable migration of the promoting elements during the reaction and decoration of iron carbide nanoparticles with the promoters. The promoters seem to facilitate CO dissociation by removing O atoms from iron carbide.

KEYWORDS: iron, olefins, Fischer–Tropsch, promoter



INTRODUCTION

Light C_2 – C_4 olefins are basic feedstocks for the production of polymers, chemical intermediates, and solvents. In industry, light olefins are mostly produced by thermal or catalytic cracking of shale gas or petroleum fractions. High-temperature Fischer–Tropsch (FT) synthesis which converts syngas derived from fossil and renewable feedstocks represents a promising and sustainable alternative for direct light olefin synthesis.^{1,2}

Fe-based catalysts are the catalysts of choice for direct light olefin synthesis from syngas. Electronic and structural promoters have been intensely used in order to increase iron dispersion, iron carbidization, FT reaction rates, and light olefin selectivity over Fe-based catalysts.^{3–7} It was suggested that alkali ions could enhance carbon monoxide dissociation because of their electron-donating effect on iron. These promoters reduce catalyst hydrogenation activity and increase olefin to paraffin ratio in the range of light hydrocarbons.^{8–11} In addition, alkali promotion leads to a higher chain growth probability and higher selectivity to the C_{5+} hydrocarbons. Transition metals such as Cu^{7,12–14} and Mn^{15,16} have been also proposed as promoters for light olefin selectivity. More recently, de Jong^{17,18} et al. reported high olefin selectivity in FT synthesis with low carbon deposition rate over iron catalysts simultaneously promoted with sodium and sulfur. The relatively low activity of the promoted catalysts and harsh reaction conditions remain a major challenge for larger scale

application of those catalysts. Thus, the development of new Fe catalysts which would be able to provide high activity and selectivity to light olefins under milder reaction conditions remains a formidable challenge.

A study of the effect of numerous promoters over Fe catalysts led us to the metals Bi and Pb. There are scarcely any reports about using these elements as promoters for iron FT catalysts. These metals are typically used as solders in plumbing, electronics, metalwork, and other applications. We report here that the promotion of iron catalysts with Bi and Pb leads to an exceptional increase in FT reaction rate (up to 10 times) and an increase in the selectivity to light olefins (up to 60%). In addition, FT synthesis can be conducted under very mild conditions and even at atmospheric pressure.

These promoting metals are crucial for the enhancement of the selectivity and activity due to several factors. First, these metals have melting points at temperatures lower ($T_{Pb} = 327$ °C; $T_{Bi} = 271$ °C) than the temperature of FT reactions (~ 350 °C). This leads to a pseudoliquid state of the promoters under the reaction conditions, their migration, and close contact between Fe and the promoters. The second important feature of these metals is their several oxidation states and easy redox

Received: April 23, 2017

Revised: July 6, 2017

Published: August 7, 2017

cycles between metal and oxide. Our results suggest that these promoters may facilitate carbon monoxide dissociation on the surface of the Fe catalyst promoted with Bi and Pb by O scavenging (Figure 1).

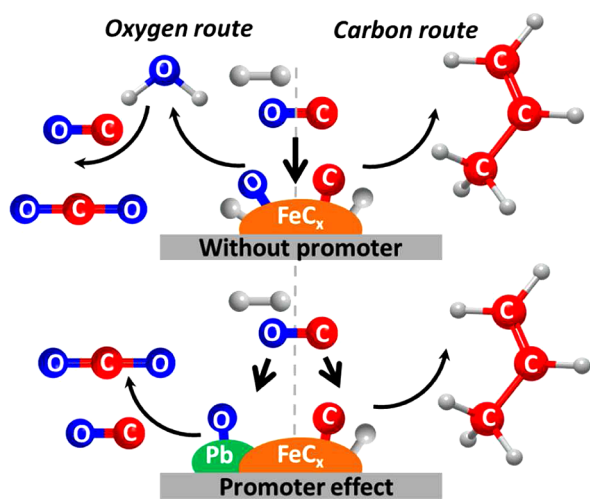


Figure 1. Effect of promotion with soldering atoms on FT synthesis.

EXPERIMENTAL SECTION

Commercial amorphous silica (CARIACT Q-10, Fuji Silysia) was used as the catalytic support. The Fe/SiO₂ and FeM/SiO₂ catalysts were prepared by incipient wetness impregnation of the support with aqueous solutions of hydrous iron nitrate (Fe(NO₃)₃·9H₂O), nitrate of bismuth (Bi(NO₃)₃·5H₂O), or nitrate of lead (Pb(NO₃)₂). The concentrations of the impregnating solutions were calculated to obtain about 10 wt % iron in the final catalysts with a ratio of Fe to promoter (M) of 100:2. After the impregnation, the catalysts were dried overnight in an oven at 100 °C. Then they were calcined in air at 400 °C for 6 h with 1 °C/min temperature ramping. The metal contents in the samples before and after reaction have been determined by X-ray fluorescence (XRF) analysis (Table S1 in the Supporting Information).

The ex situ X-ray powder diffraction (XRD) experiments were conducted using a Bruker AXS D8 diffractometer with Cu K α radiation ($\lambda = 0.1538$ nm). The XRD patterns were collected in the 20–70° (2 θ) range. The identification was carried out by comparison with JCPDF standard spectra software. The average crystallite size of Fe₂O₃ or iron carbides was calculated using the diffraction peaks according to the Scherrer equation.

The SSITKA apparatus has been described in detail in our previous report.¹⁹ It is composed of two independent gas feed lines. The first line is dedicated to unlabeled compounds and a tracer (CO, H₂, He, and Ne) and the second line to the isotope-labeled compounds (e.g., ¹³CO). The experiment consists of the sudden replacement of one reactant by its labeled counterpart without any modification of the reaction total pressure, flow, and reactant chemical composition, while an inert tracer is also abruptly removed from the feed. Further details about the SSITKA method are available elsewhere.^{20–23} The incorporation of the labeled atom into the reaction products was continuously monitored. The switches between the two gas lines were realized using a two-position four-way Valco valve. The outlet stream was monitored with a QMG 432

Omnistar mass spectrometer. The maintenance of isobaric, isoflow, and isothermal reaction conditions during the switch guaranteed the quasi-steady-state conditions of the catalyst.

The kinetics of oxygen removal via reaction with carbon monoxide has been studied in the SSITKA unit after carburization of 100 mg of the catalyst in 5 mL/min of CO at 350 °C over 10 h. Afterward the reactor was cooled to 300 °C and purged with 2 mL/min of nitrogen. Then 100 mg of the catalysts was exposed to 0.05 mL of water at 300 °C to generate oxygen species on the catalyst surface with subsequent flush of the catalyst under a flow of nitrogen. The water-gas shift (WGS) reaction on the surface of the catalyst was performed by exposure of the catalyst to a CO flow of 5 mL/min. The signal of CO₂ was registered with a mass spectrometer.

In situ XPS spectra were recorded using a Physical Electronics Quantum 2000 Scanning ESCA Microprobe spectrometer using an Al K α source (1486.6 eV). The powdered samples were made into 6 mm diameter pellets. In each experiment, the XPS spectra of the fresh catalyst were first measured and then the pellets were transferred into the in situ reaction cell heated under a flow of CO (50 mL/min, 1 bar) from room temperature to 350 °C at a heating rate of 10 °C/min and kept for 3 h. The treated sample was then transferred under vacuum to the analytical chamber to record the XPS spectra of the catalyst. After that, the catalyst was exposed to a H₂/CO mixture (50 mL/min, 1 bar) at 350 °C for 3 h in the reaction cell and transferred again under vacuum to the analytical chamber to record the XPS spectra. The binding energies were corrected with respect to C 1s at 284.6 eV.

The TEM observation of the samples was performed on a FEI Tecnai F30 electron microscope, operating at 300 kV. Prior to the analysis, the samples were dispersed by ultrasound in ethanol solution for 5 min, and a drop of the solution was deposited onto a carbon membrane onto a 400 mesh copper grid. The HADDF-STEM images were acquired in scanning transmission electron microscopy (STEM) mode. The microscope was also equipped with an EDX analyzer from Oxford Instruments.

The magnetic properties of the catalysts were investigated in situ using a Foner vibrating-sample magnetometer^{24,25} with 20 mg catalyst loading. The magnetometer was calibrated using 1 mg of pure metallic Fe before each experiment. First, under a 15 mL/min feed of pure CO, the sample was heated to 200 °C with a 6.6 °C/min ramping and kept for 10 min at this temperature and then sequentially heated to 350 °C with a 4.7 °C/min ramping and kept for 120 min. After the activation the sample was cooled to room temperature under a flow of CO. During the whole treatment, the saturation magnetization curve was recorded by the magnetometer.

Carbon monoxide hydrogenation was carried out using a Flowrence high-throughput unit (Avantium) equipped with 16 parallel milli-fixed-bed reactors ($d = 2$ mm) operating at a pressure range of 1–20 bar, H₂/CO molar ratio of 1, and GHSV from 1.5 to 27 L h⁻¹ g_{cat}⁻¹. The catalyst loading was 100 mg per reactor. The catalysts were pressed and sieved to particles of sizes 50–150 μ m. Previous reports²⁶ suggest the absence of intraparticle diffusion limitations under these conditions. The reactor was a stainless steel tube with inner diameter of 2.0 mm and length of 15 cm. On the top and on the bottom the reactor was filled with the layers of inert SiC. Prior to the catalytic test all of the samples were activated under a flow of CO at atmospheric pressure over 10 h at 350 °C. During the activation step, the temperature ramp was 3 °C/

Table 1. Catalyst Characterization and Catalytic Performance ($H_2/CO = 1$) after 60 h of Reaction

catalyst	reaction conditions			activity			selectivity				
	$T, ^\circ C$	$P, \text{ bar}$	$GHSV, L\ g^{-1}\ h^{-1}$	$FTY, 10^{-4}\ \text{mol}\ g_{Fe}^{-1}\ s^{-1}$	$X_{CO}, \%$	$S_{CO_2}, \%$	product distribn (% C_{at} , CO_2 free)				
							$S_{CH_4}, \%$	$S_{C_2-C_4}, \%$	$S_{C_2-C_4}, \%$	$S_{C_5+}, \%$	α
Fe/SiO ₂	350	10	1.5	0.3	32	43	15	22	22	40	0.5
	350	10	3.4	0.4	16	35	14	33	10	43	0.48
	350	10	4.5	0.4	13	34	15	41	11	33	0.48
FeBi/SiO ₂	350	10	13.4	1.8	21	42	27	36	16	21	0.46
	350	10	6.7	2.4	58	47	24	32	17	27	0.47
	350	10	3.4	1.8	75	49	24	26	21	29	0.48
	350	1	3.4	0.4	17	48	29	53	8	10	0.35
FePb/SiO ₂	350	10	3.4	1.3	60	45	20	31	11	38	0.48
	350	1	3.4	1.2	55	45	22	32	12	34	0.39
	250	5	3.4	0.1	6	10	15	26	9	50	0.54
Fe/SiO ₂ + Pb/SiO ₂ ^a	350	10	3.4	0.7	34	42	27	39	16	18	0.44
Fe/SiO ₂ + Bi/SiO ₂ ^a	350	10	3.4	0.8	38	42	29	38	12	21	0.42

^aMechanical mixture with the same content of iron and promoter as in the composite catalysts.

min. After the reduction, the catalysts were cooled to 180 °C and a flow of premixed syngas was gradually introduced to the catalysts. When the pressure attained the pressure of the reaction, the temperature was slowly increased to the reaction temperature. The gaseous reaction products were analyzed by online gas chromatography. Analysis of permanent gases was performed using a molecular sieve column and a thermal conductivity detector. Carbon dioxide and C_1 – C_4 hydrocarbons were separated in a PPQ column and analyzed by a thermoconductivity detector. The C_5 – C_{12} hydrocarbons were analyzed using a CP-Sil5 column and a flame-ionization detector. High-molecular-weight products were collected at atmospheric pressure in vials heated at 80 °C. The carbon monoxide contained 5% of helium, which was used as an internal standard for calculating carbon monoxide conversion. The relative standard deviation for CO conversion during the reaction test with the same catalyst did not exceed 5%. Iron time yields (FTY) were expressed as moles of CO converted per gram of total iron per second. The product conversions and selectivities were reported as the percentage of CO converted and as percentage of CO converted into a given product, respectively, and expressed on a carbon basis:

$$\text{conversion} = (\text{CO to the reactor (mol/s)})$$

$$- \text{CO (mol/s) from the reactor})$$

$$/((\text{CO to the reactor (mol/s)}) \times 100\%$$

$$\text{selectivity to } CO_2 = (\text{C atoms of } CO_2 \text{ from the reactor (mol/s)}) / ((\text{CO to the reactor (mol/s)} - \text{CO (mol/s) from the reactor}) \times 100\%$$

$$\text{selectivity (} CO_2 \text{ free)} = (\text{C atoms of product from the reactor}) / ((\text{CO to the reactor (mol/s)} - \text{CO out of the reactor (mol/s)} - \text{C atoms of } CO_2 \text{ from the reactor}) \times 100\%$$

The carbon balance for C_1 – C_{12} in all tests was in the range 85–95%, depending on the catalyst and conversion.

The chain growth probability has been determined as the slope of the plot of the logarithm of the product distribution as

a function of its chain length for the C_5 – C_{12} hydrocarbon range:

$$\log(W_n/n) = n \log \alpha + \text{const}$$

The static hydrogenation tests of the catalysts have been conducted in the same unit at 300 °C, 20 bar, and H_2 flow with $GHSV = 0.75\ L\ g^{-1}\ h^{-1}$ after carburization pretreatment.

RESULTS AND DISCUSSION

Enhancement in Catalytic Performance on Promotion. Methane, light C_2 – C_4 olefins, light C_2 – C_4 paraffins, C_5 – C_{10} hydrocarbons, carbon dioxide, and water were the principal products of carbon monoxide hydrogenation under the studied operating conditions. The catalysts were tested under a wide range of reaction pressures (1–20 bar). Iron time yield (FTY) increased 5–10 times on promotion (Table 1). Interestingly, Fe catalysts promoted by the soldering metals were active even at low pressures in comparison with Fe/SiO₂. Figure 2 shows carbon monoxide conversion obtained on these catalysts at iso-GHSV but at different total pressures. As expected for the reference silica-supported iron catalyst, carbon monoxide

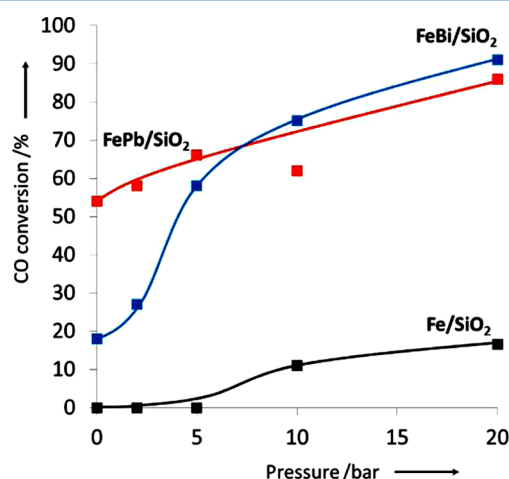


Figure 2. Carbon monoxide conversion on iron catalysts as a function of the reaction total pressure. Reaction conditions: $T = 350\ ^\circ C$, $H_2/CO = 1$, $GHSV = 3.4\ L\ g^{-1}\ h^{-1}$. The data were obtained by increasing pressure in the row 0–2–5–10–20 bar.

conversion strongly decreases with a decrease in total pressure. There was already no conversion at 5 bar, which means that the unpromoted iron catalyst cannot be used for FT synthesis at low pressure.

Surprisingly, the Bi- and Pb-promoted catalysts demonstrated significant activity even at atmospheric pressure. This is also indicative of significant modifications in the reaction kinetics and possibly mechanism of FT synthesis. Note that the FTY observed over the Bi- and Pb-promoted Fe/SiO₂ (FTY = 2.4×10^{-4} mol g_{Fe}⁻¹ s⁻¹) catalysts was comparable with those for the most active iron catalytic systems known so far^{1,2,27,28} (FTY, Table 1). For example, highly dispersed Fe-based catalysts synthesized by decomposition of MOF provide an FTY of 4.4×10^{-4} mol g_{Fe}⁻¹ s⁻¹ under comparable reaction conditions.²⁹ In the presence of promoters, FT synthesis might be performed under much milder conditions and even at atmospheric pressure (Table 1 and Figure 2) and lower temperature (Figure S1 in the Supporting Information). Interestingly, the promoted catalysts demonstrate a continuous increase in the catalytic activity during the reaction (Figure 3). Note that the unpromoted Fe/SiO₂ deactivates under the same conditions.

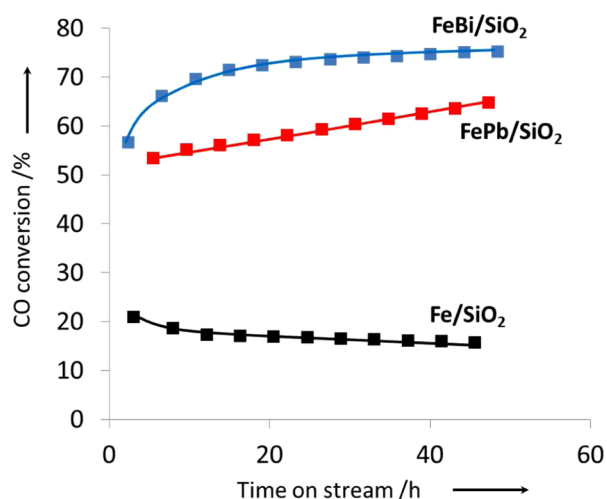
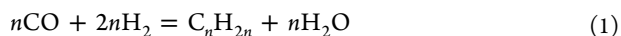


Figure 3. Effect of the type of promoter on the CO conversion as a function of time. Reaction conditions: $P = 10$ bar, $H_2/CO = 1$, $GHSV = 3.4$ L g⁻¹ h⁻¹.

The selectivity data for the Bi-promoted iron catalyst and Fe/SiO₂ measured as a function of carbon monoxide conversion are summarized in Figures 4 and 5 and Figures S2 and S3 in the Supporting Information. Similar catalytic performance has been observed for the lead-promoted catalyst (Table 1 and Figure S4 in the Supporting Information). The reactions occurring during FT synthesis over iron catalysts are



For the Fe/SiO₂ catalyst, carbon dioxide selectivity increases with carbon monoxide conversion, moving to the stoichiometric value of 50% (Figure 4 and eq 3). This is consistent with the hypothesis that carbon dioxide can be also produced by a WGS, which is a secondary reaction of FT synthesis (eq 2). Note, however, that over the Bi- and Pb-promoted catalysts

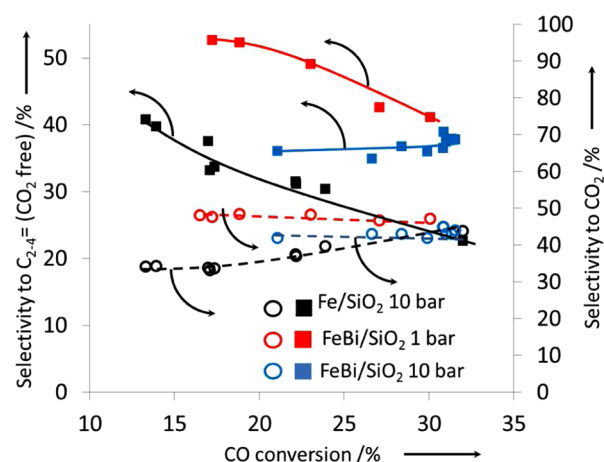


Figure 4. Effect of Bi promoter on the selectivity versus CO conversion. Reaction conditions: $T = 350$ °C, $H_2/CO = 1$, $P = 1$ and 10 bar for FeBi/SiO₂ and Fe/SiO₂, respectively, $GHSV = 1.5$ – 27 L g⁻¹ h⁻¹.

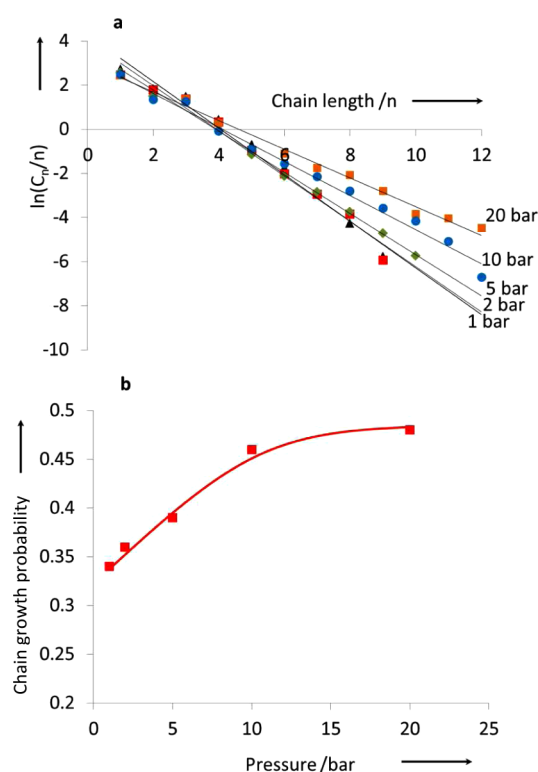


Figure 5. (a) ASF plots and (b) chain growth probabilities over FeBi/SiO₂ catalyst at different pressures. Reaction conditions: $P = 1$ – 20 bar, $H_2/CO = 1$, $GHSV = 1.5$ – 3.4 L g⁻¹ h⁻¹, $T = 350$ °C. The conversion was about 30%.

high carbon dioxide selectivity is observed even at very low carbon monoxide conversion levels (Figure 4 and Table 1). This suggests that, on these catalysts, the primary route of carbon dioxide formation might take place together with the secondary WGS reaction occurring through reaction of carbon monoxide with water.

Note also that the selectivity to light olefins decreases with an increase in CO conversion over Fe/SiO₂ catalyst, which is probably due to secondary hydrogenation of olefins to alkanes³⁰ (Table 1 and Figure 4). The presence of promoters stabilizes the selectivity to light olefins close to 35% at 10 bar at

high conversions with suppression of the secondary transformation of olefins (Figure S2 in the Supporting Information). This effect is similar to those observed in the presence of alkali promoters. Differently from alkali promotion, the Bi- and Pb-promoted catalysts exhibit lower selectivity to long-chain hydrocarbons. High selectivity to long-chain hydrocarbons limits the selectivity to light olefins over alkali-promoted catalysts. At lower reaction pressures (1 bar) the selectivity to light olefins can be further increased by about 20%, reaching 53% at conversions of 17%, which corresponds to the best published reports^{1,2,9,27,28} (Table 1, Figure 4, and Figure S3 in the Supporting Information). Importantly, the increase in selectivity to light olefins at lower reaction pressure coincides with the decrease in the selectivity to the C_{5+} hydrocarbons (Figure S3). This effect is the result of the gradual decrease in the chain growth probability (α) with the pressure decrease (Figure 5). The Anderson–Schulz–Flory distribution predicts maximum selectivities to C_2 – C_4 hydrocarbons of about 50% at α values between 0.3 and 0.6.³¹ The pressure decrease from 10 to 1 bar leads to a decrease in α from 0.5 to 0.35, which is favorable for higher olefin selectivity. It is interesting to note that an increase in the CO conversion by lowering the GHSV at atmospheric pressure leads to a gradual decrease in the selectivity to light olefins (Figure 4 and Figure S3). Note that the selectivity to olefins decreases more slowly with conversion at higher pressure. The possible explanation of this effect could be in reinsertion of light olefins in the growing chains at higher CO conversion at low pressure, leading to a decrease in the selectivity to light olefins and increase in the selectivity to C_{5+} hydrocarbons. Thus, use of soldering metal promoters makes it possible to produce light olefins over iron catalysts with high yield and selectivity. Note that the selectivity gain is more pronounced at lower reaction pressure.

Mechanism of the Promotion. Previous reports suggest^{1,2,8,12,32} that FT reaction on iron catalyst occurs on the surface of iron carbides. The FT reaction rate on the iron catalysts could be therefore a function of the number and intrinsic reactivity (turnover frequency) of iron carbide surface sites.³² The concentration of iron carbide surface sites is a function of iron carbide dispersion and extent of carbidization. Investigation of the calcined catalysts by XRD (Figure 6) did not reveal any effect of the promoters on the dispersion of supported hematite iron oxide. The size of iron oxide particles was always between 15 and 17 nm, and it was not much affected by the promoters. Figure S5 in the Supporting Information displays XRD patterns of the unpromoted and Bi- and Pb-promoted iron catalysts after the FT reaction tests. Very broad XRD peaks attributed to an iron carbide phase has been detected. Again, the width of the iron carbide XRD peak has also been not much affected by the promotion. The sizes of iron carbide nanoparticles according to XRD were in the range from 4 to 6 nm (Table S1 in the Supporting Information). This suggests that iron carbide dispersion is also not affected by the promotion with soldering metals.

An in situ magnetic method (Figure 6) did not either uncover any effect of the promoters on the extent of carbidization and chemical composition of iron carbide phases. Prior to the magnetic measurements, the catalysts were treated in CO at 350 °C using a procedure similar to that used for catalyst activation. Catalyst magnetization was zero at temperatures higher than the Curie temperature of Hägg iron carbide. Magnetite has a Curie temperature at 585 °C and (if present in the catalysts) must be detected by a magnetic method. Thus,

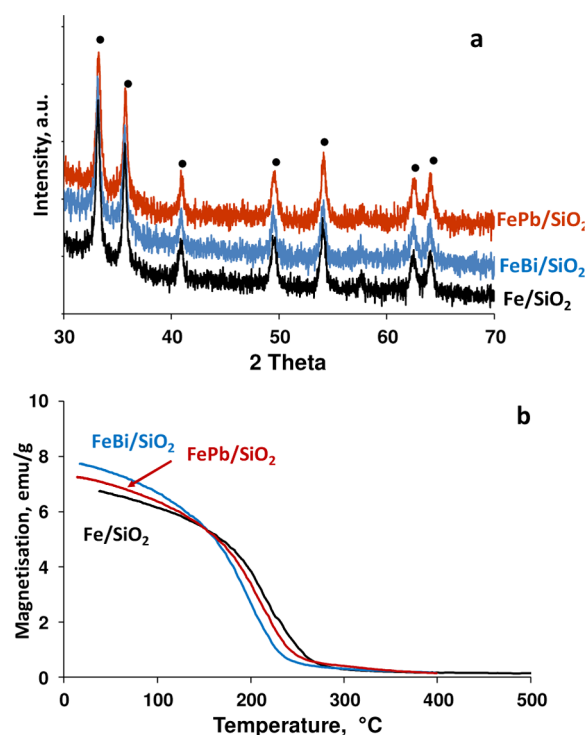


Figure 6. (a) XRD patterns of the catalysts after calcination and (b) catalyst magnetization during cool-down after CO treatment at 350 °C.

the observed zero magnetization at $T > 250$ °C suggests complete iron carbidization and, thus, extremely low concentration of magnetite in the activated catalysts. In addition, catalyst magnetizations at room temperature in the catalysts carbidized in CO at 350 °C were the same for unpromoted and promoted catalysts. This is indicative of the same concentration of iron carbide. The Curie temperature (205–220 °C) measured in both unpromoted and promoted catalysts corresponded to the Hägg iron carbide (χ -Fe₅C₂).

The catalytic phenomena over the Bi- and Pb-promoted catalysts cannot therefore be attributed to changes in the iron carbide dispersion, amount, and composition. Thus, the performance of the promoted catalysts seems to be affected not by variation of the number of active sites but by the modification of their intrinsic activity (turnover frequency).

The reactivity of iron carbide in the promoted and unpromoted catalysts has been characterized by hydrogenation under the reaction conditions. Our earlier studies³³ indicated that the catalysts with higher rates of iron carbide hydrogenation exhibited high activity in FT synthesis. Comparable hydrogenation profiles were observed with unpromoted and promoted catalysts. The results suggest no significant difference in the concentration and reactivity of the iron carbide species in the unpromoted and promoted catalysts in the presence of hydrogen (Figure S6 in the Supporting Information).

The observed strong effects of Bi and Pb promoters on the catalytic performance of Fe catalysts might be due to the intimate contact between Fe and promoter. It is expected that low melting point of soldering metals can lead to their higher mobility. In order to prove this, we have prepared mechanical mixtures of Fe/SiO₂ with either Bi/SiO₂ or Pb/SiO₂ and performed catalytic tests (Table 1 and Figure 7). CO conversion over mechanical mixtures was much higher than that over the unpromoted Fe/SiO₂ and comparable with the

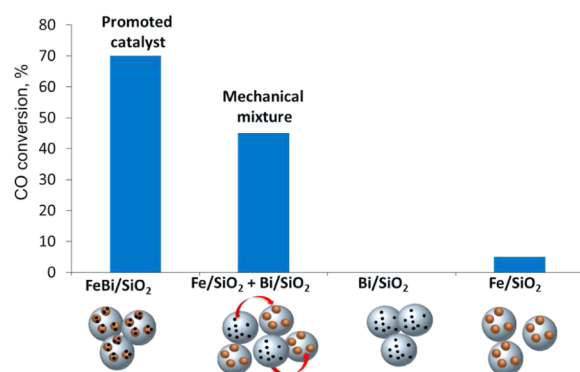


Figure 7. Conversion of CO over FeBi/SiO₂, Bi/SiO₂, Fe/SiO₂, and mechanical mixture Fe/SiO₂ + Bi/SiO₂. Reaction conditions: H₂/CO = 1, *P* = 10 bar, *T* = 350 °C, GHSV = 3.4 L g_{cat}^{−1} h^{−1}, TOS = 100 h.

conversion over the coimpregnated promoted catalysts. This is indicative of substantial migration of the promoters during the reaction.

The conducted TEM-EDX confirms this assumption (Figure 8). Amorphous silica consists of globules of size 5–10 nm

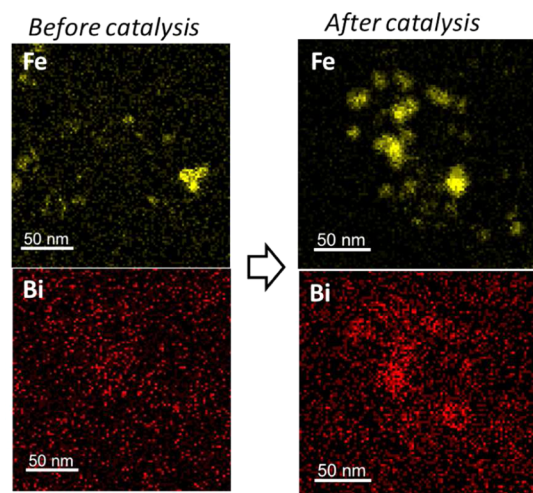


Figure 8. TEM-EDX images for Fe and Bi before after catalysis for the mechanical mixture Fe/SiO₂ + Bi/SiO₂ (see Figure 7).

(Figure S7 in the Supporting Information). In the well mechanically mixed sample, TEM-EDX detected both Fe nanoparticles in the range 10–40 nm and Bi particles which were much smaller because of lower metal content. Note that, in the initial mechanical mixtures, Bi was uniformly distributed with the same density in all parts of the image (Figure 8 and Figure S8 in the Supporting Information). After FT synthesis was conducted, the EDX maps clearly showed preferential Bi localization in the close proximity of iron carbide nanoparticles (Figure 8 and Figure S9 in the Supporting Information). The significant increase in the density of Bi in the interface with Fe nanoparticles might be explained by migration of Bi under the reaction conditions and their interaction with iron. Intensive migration of promoter could also lead to changes in the composition of the catalyst due to partial deposition of promoter over the walls of the reactor. However, analysis of the catalyst composition before and after catalysis (Table S1 in the Supporting Information) shows that no promoter was lost from the catalyst during the reaction

FT synthesis is a multistage catalytic reaction. Previous reports suggest that FT synthesis over iron catalysts may proceed with the Mars–Van Kravlen sequence.^{34,35} One of the initial steps of FT synthesis on iron catalysts is carbon monoxide dissociative adsorption. Carbon monoxide dissociation over iron catalysts can be either direct or hydrogen-assisted. A recent DFT report³⁶ showed that, differently from cobalt catalysts, direct carbon monoxide dissociation proceeds much more easily on iron catalysts in comparison to the hydrogen-assisted process. Carbon monoxide direct dissociation results in formation of surface carbide and chemisorbed oxygen species followed by oxygen removal via its reaction with either hydrogen or carbon monoxide. Chemisorbed carbon can be then hydrogenated to CH_x monomer, which then can be involved in the FT surface polymerization. It can be expected that the rate of these elementary steps can be affected differently by the promoting elements. First, the rate of carbon monoxide adsorption and dissociation on the unpromoted and Bi- and Pb-promoted iron catalysts in the presence of syngas and at the reaction temperature (350 °C) was evaluated using SSITKA experiments.¹⁹ During the switches from ¹²CO/H₂/Ne to ¹³CO/H₂, the ¹²CO carbon monoxide response curves were indistinguishable from the transient response of the inert gas (Figure S10 in the Supporting Information). This suggests that the rates of carbon monoxide adsorption and desorption are very fast on both unpromoted and promoted iron catalysts. Importantly, no molecular adsorption of carbon monoxide was observed on iron carbide under the reaction conditions. This also suggests that the only type of CO strong adsorption on iron catalysts can be dissociative.

The rates of oxygen removal produced on CO dissociation on the Bi- and Pb-promoted and unpromoted iron catalysts were evaluated by the reaction with carbon monoxide. The unpromoted and Bi- and Pb-promoted catalysts were first carbided in carbon monoxide under typical conditions used for the catalyst activation. Then the catalysts were exposed to water to generate oxygen species on the catalyst surface. During the contact of the catalyst with water, hydrogen formation has been observed (Figure S11 in the Supporting Information). Hydrogen formation is probably the result of the oxidation of iron carbide with water. Interestingly, the hydrogen production rate and oxidation of iron carbide in the unpromoted catalyst were slow. At the same time, over the promoted catalysts intense and rapid hydrogen formation has been observed at lower reaction time.

After this treatment, the catalysts were again exposed to carbon monoxide to evaluate the reactivity of the generated oxygen species in the presence of carbon monoxide. Carbon dioxide production rates from CO after partial oxidation of the carbide surface in water are displayed in Figure 9. Interestingly, the rates of carbon dioxide production were much higher over the Pb-promoted and in particular Bi-promoted catalysts in comparison to the unpromoted counterpart. The trend was similar to hydrogen rate formation during catalyst oxidation by water. This suggests that the rates of oxygen scavenging and its removal via its reaction with carbon monoxide are significantly enhanced in the presence of promoters. One of the plausible explanations is displayed in Figure 1. It is relevant to the enhancement of the diffusion of the oxygen formed on iron carbide to the promoting atoms (Bi or Pb) situated in close proximity. Oxygen migration to the promoters leads to their partial oxidation (Figure 1). Oxygen species will then be removed much more easily from the promoter by CO than

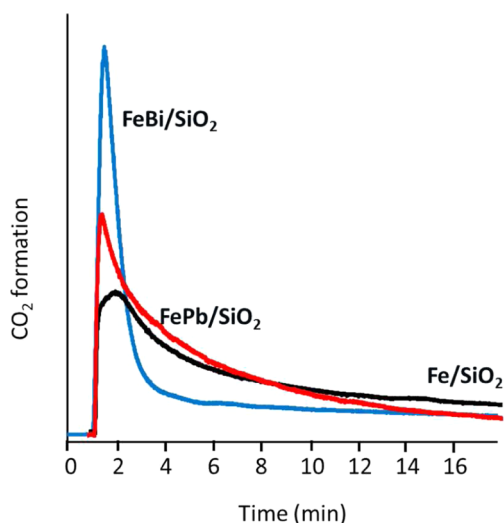


Figure 9. Rate of carbon dioxide production after exposure of the activated silica supported iron catalysts pretreated with water to CO at 300 °C.

those located on the surface of iron carbide. The observed trend is consistent with the electrochemical potentials of iron ($\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe(s)}, -0.44 \text{ V}$), bismuth ($\text{Bi}^{3+} + 3\text{e}^- \rightleftharpoons \text{Bi(s)}, +0.308 \text{ V}$), and lead ($\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb(s)}, -0.126 \text{ V}$). Oxygen removal by reaction with CO leading to the reduction of the promoter will be thermodynamically favored on Bi and Pb in comparison to Fe.

This suggestion is also consistent with the XPS characterization data. The Pb-promoted and unpromoted iron catalysts were activated in carbon monoxide and exposed to syngas in the pretreatment chamber of an XPS spectrometer. For XPS analysis, the catalyst was transferred without exposure to air from the pretreatment chamber to the XPS analysis chamber (Figure 10). Note that, before the reaction, Pb was in the oxidized state. In the catalyst activated in carbon monoxide, Pb was mostly observed in the metallic state. This suggests that the catalyst treatment with carbon monoxide results in the removal

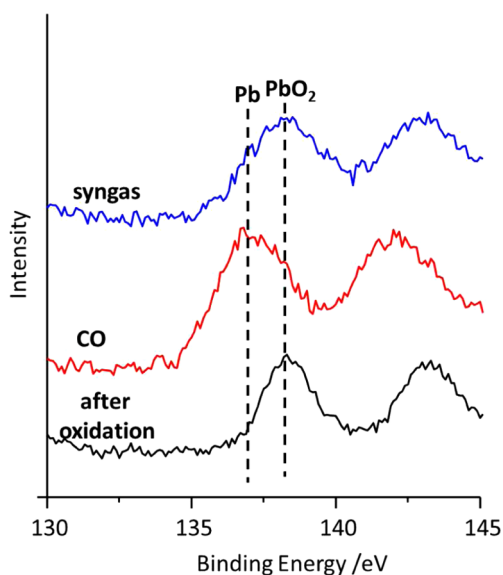


Figure 10. Pb 4f XPS spectra of the FePb/SiO₂ catalyst after calcination and exposure to carbon monoxide and syngas.

of all oxygen atoms from the promoter and its reduction to the metallic state. Interestingly, noticeable modifications of the XPS spectra of lead were observed after the exposure of the activated catalyst to syngas. The presence of syngas resulted in lead oxidation. This is consistent with the hypothesis about scavenging of oxygen atoms formed on carbon monoxide dissociation on iron carbide by the promoter, followed by the removal of oxygen species with carbon monoxide. This suggestion agrees with high CO₂ selectivity at low carbon monoxide conversion observed on the promoted catalysts, indicating that CO₂ could be a primary reaction product over the Bi- and Pb-promoted catalysts (Figure 4).

CONCLUSION

To conclude, we uncovered a very strong promoting effect of iron catalysts with soldering metals (Bi, Pb) on high-temperature FT synthesis. The FT rate and light olefin selectivity increase by 5–10 times and up to 60%, respectively, in comparison to the unpromoted catalyst. The Bi- and Pb-promoted catalysts also provide an opportunity to selectively produce light olefins from syngas with high yields at atmospheric pressure. The intrinsic activity of iron carbide active sites seems to be enhanced by the promoters, which are localized in close proximity to the iron carbide nanoparticles and can facilitate CO dissociation by O removal.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.7b01307.

Catalytic results, XRD of carbidized samples, methane formation rate during hydrogenation, TEM image of Fe/SiO₂, SSITKA, and H₂ production during water treatment (PDF)

AUTHOR INFORMATION

Corresponding Author

*A.Y.K.: tel, +33 3 20 33 54 39; fax, +33 3 20 43 69 53; e-mail, andrei.khodakov@univ-lille1.fr.

ORCID

Vitaly V. Ordonsky: 0000-0002-4814-5052

Andrei Y. Khodakov: 0000-0003-4599-3969

Present Address

^{||}Eco-Efficient Products and Processes Laboratory (E2P2L), UMI 3464 CNRS-Solvay, Shanghai, P.R. China

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

B.G. and A.C. are thankful to the Chinese Scholarship Council and Brazilian CAPES/COFECUB Foundation, respectively, for providing them Ph.D. stipends. The authors acknowledge financial support of the French National Research Agency (Projects DirectSynBioFuel, ANR-15-CE06-0004 and NANO4-FUT, ANR-16-CE06-0013). The REALCAT platform benefits from a Governmental subvention administrated by the French National Research Agency (ANR) within the framework of the “Future Investments” program (PIA), with contractual reference “ANR-11-EQPX-0037”.

■ REFERENCES

- (1) Torres Galvis, H. M.; de Jong, K. P. *ACS Catal.* **2013**, *3*, 2130–2149.
- (2) Davis, B. H. *Ind. Eng. Chem. Res.* **2007**, *46*, 8938–894.
- (3) Ngantsoue-Hoc, W.; Zhang, Y. Q.; O'Brien, R. J.; Luo, M. S.; Davis, B. H. *Appl. Catal., A* **2002**, *236* (1–2), 77–89.
- (4) Lohitharn, N.; Goodwin, J. G. *J. Catal.* **2008**, *260*, 7–16.
- (5) Lohitharn, N.; Goodwin, J. G., Jr.; Lotero, E. *J. Catal.* **2008**, *255*, 104–113.
- (6) Luo, M. S.; Davis, B. H. *Appl. Catal., A* **2003**, *246*, 171–181.
- (7) Wan, H.; Wu, B.; Zhang, C.; Xiang, H.; Li, Y. *J. Mol. Catal. A: Chem.* **2008**, *283*, 33–34.
- (8) Dictor, R. A.; Bell, A. T. *J. Catal.* **1986**, *97* (1), 121–136.
- (9) Bukur, D. B.; Mukesh, D.; Patel, S. A. *Ind. Eng. Chem. Res.* **1990**, *29* (2), 194–204.
- (10) Ribeiro, M. C.; Jacobs, G.; Davis, B. H.; Cronauer, D. C.; Kropf, A. J.; Marshall, C. L. *J. Phys. Chem. C* **2010**, *114* (17), 7895–7903.
- (11) Raje, A. P.; O'Brien, R. J.; Davis, B. H. *J. Catal.* **1998**, *180* (1), 36–4.
- (12) Pansanga, K.; Lohitharn, N.; Chien, A. C. Y.; Lotero, E.; Panpranot, J.; Praserttham, P.; Goodwin, J. G., Jr. *Appl. Catal., A* **2007**, *332* (1), 130–137.
- (13) Zhang, C.-H.; Yang, Y.; Teng, B.-T.; Li, T.-Z.; Zheng, H.-Y.; Xiang, H.-W.; Li, Y.-W. *J. Catal.* **2006**, *237* (2), 405–415.
- (14) Al-Dossary, M.; Fierro, J. L. G.; Spivey, J. J. *Ind. Eng. Chem. Res.* **2015**, *54* (3), 911–921.
- (15) Campos, A.; Lohitharn, N.; Roy, A.; Lotero, E.; Goodwin, J. G., Jr.; Spivey, J. J. *Appl. Catal., A* **2010**, *375* (1), 12–16.
- (16) Lohitharn, N.; Goodwin, J. G., Jr.; Lotero, E. *J. Catal.* **2008**, *255* (1), 104–113.
- (17) Torres Galvis, H. M.; Bitter, J. H.; Khare, C. B.; Ruitenbeek, M.; Dugulan, A. I.; de Jong, K. P. *Science* **2012**, *335* (6070), 835–838.
- (18) Koeken, A. C. J.; Torres Galvis, H. M.; Davidian, T.; Ruitenbeek, M.; de Jong, K. P. *Angew. Chem., Int. Ed.* **2012**, *51*, 7190–7193.
- (19) Legras, B. V.V.; Ordonsky, C.; Dujardin, M.; Virginie, A. Y.; Khodakov, A. *ACS Catal.* **2014**, *4*, 2785.
- (20) Happel, J. *Chem. Eng. Sci.* **1978**, *33*, 1567.
- (21) Biloen, P. *J. Mol. Catal.* **1983**, *21*, 17.
- (22) Biloen, P.; Helle, J. N.; van den Berg, F. G. A.; Sachtler, W. M. H. *J. Catal.* **1983**, *81*, 450.
- (23) Shannon, S. L.; Goodwin, J. G. *Chem. Rev.* **1995**, *95*, 677.
- (24) Chernavskii, P. A.; Dalmon, J.-A.; Perov, N. S.; Khodakov, A. Y. *Oil Gas Sci. Technol.* **2009**, *64*, 25.
- (25) Chernavskii, P. A.; Khodakov, A. Y.; Pankina, G. V.; Girardon, J.-S.; Quinet, E. *Appl. Catal., A* **2006**, *306*, 108.
- (26) Post, M. F. M.; van 't Hoog, A. C.; Minderhoud, J. K.; Sie, S. T. *AIChE J.* **1989**, *35*, 1107–1114.
- (27) Xie, J.; Torres Galvis, H. M.; Koeken, A. C. J.; Kirilin, A.; Dugulan, A. I.; Ruitenbeek, M.; de Jong, K. P. *ACS Catal.* **2016**, *6*, 4017–4024.
- (28) Cheng, Y.; Lin, J.; Xu, K.; Wang, H.; Yao, X.; Pei, Y.; Yan, S.; Qiao, M.; Zong, B. *ACS Catal.* **2016**, *6*, 389–399.
- (29) Santos, V. P.; Wezendonk, T. A.; Delgado Jaen, J. J.; Dugulan, A. I.; Nasalevich, M. A.; Islam, H.-U.; Chojecki, A.; Sartipi, S.; Sun, X.; Hakeem, A. A.; Koeken, A. C. J.; Ruitenbeek, M.; Davidian, T.; Meima, G. R.; Sankar, G.; Kapteijn, F.; Makkee, M.; Gascon, J. *Nat. Commun.* **2015**, *6*, 6451.
- (30) Yang, J.; Ma, W.; Chen, D.; Holmen, A.; Davis, B. H. *Appl. Catal., A* **2014**, *470*, 250–260.
- (31) Nowicki, L.; Ledakowicz, S.; Bukur, D. *Chem. Eng. Sci.* **2001**, *56*, 1175–1180.
- (32) Torres Galvis, H. M.; Bitter, J. H.; Davidian, T.; Ruitenbeek, M.; Dugulan, A. I.; de Jong, K. P. *J. Am. Chem. Soc.* **2012**, *134*, 16207–16215.
- (33) Ordonsky, V. V.; Legras, B.; Cheng, K.; Paul, S.; Khodakov, A. Y. *Catal. Sci. Technol.* **2015**, *5*, 1433–1437.
- (34) Gracia, J. M.; Prinsloo, F. F.; Niemantsverdriet, J. W. *Catal. Lett.* **2009**, *133* (3–4), 257–261.
- (35) Ozbek, M. O.; Niemantsverdriet, J. W. *J. Catal.* **2014**, *317*, 158–166.
- (36) Pham, T. H.; Duan, X.; Qian, G.; Zhou, X.; Chen, D. *J. Phys. Chem. C* **2014**, *118*, 10170–10176.

Supporting information

Soldering of iron catalysts for direct synthesis of light olefins from syngas under mild reaction conditions

Vitaly V. Ordonsky, Yuan Luo, Bang Gu, Alexandre Carvalho, Petr A. Chernavskii,
Kang Cheng and Andrei Y. Khodakov*

*Unité de Catalyse et de Chimie du Solide, UMR 8181 CNRS, Bât. C3, Ecole Centrale
de Lille, Université de Lille, 59655 Villeneuve d'Ascq, France*

* E-mail: andrei.khodakov@univ-lille1.fr

Table S1. XRF analysis of metal content in the catalysts before and after catalysis

Samples	Size of Fe₂O₃ and Fe carbide, nm (XRD)	Fe, wt %	Bi, wt %	Pb, wt %
Fe/SiO ₂ -before	17	15.46	-	-
Fe/SiO ₂ -after	4.4	14.48	-	-
FeBi/SiO ₂ -before	15	14.43	0.94	-
FeBi/SiO ₂ -after	5.8	16.23	1.10	-
FePb/SiO ₂ -before	16	14.5	-	1.27
FePb/SiO ₂ -after	5.2	13.8	-	1.25

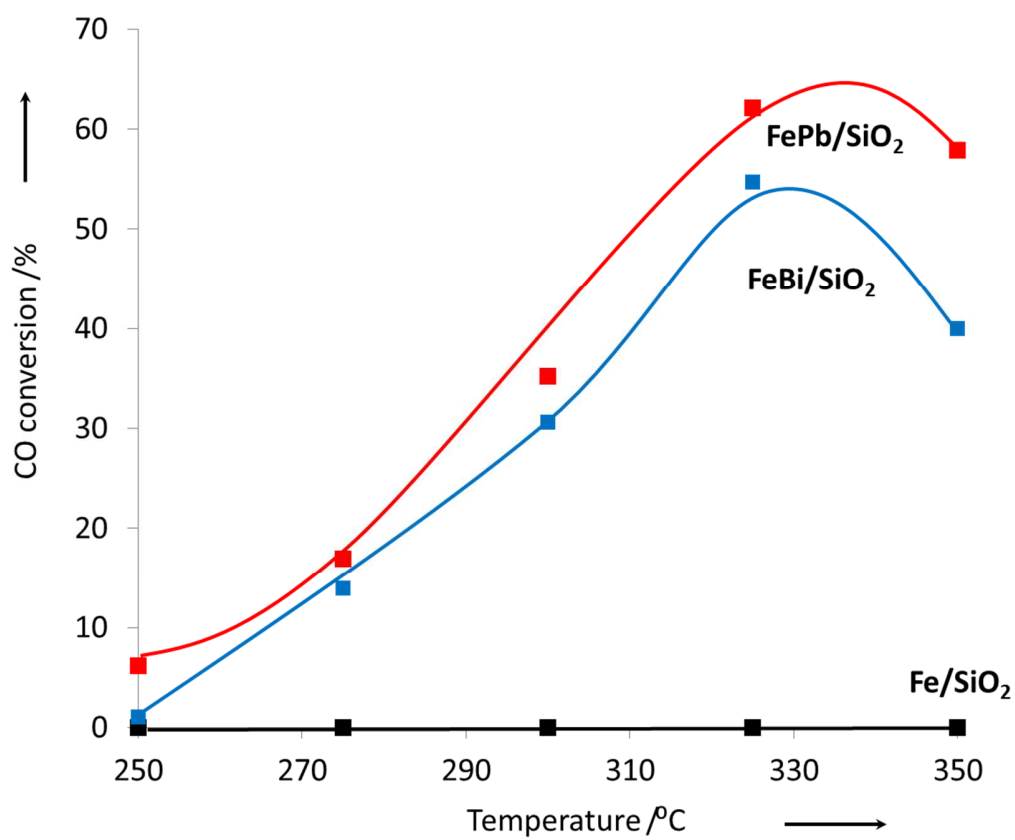


Figure S1. Effect of the type of promoter on the CO conversion at different temperatures. Reaction conditions: P= 5 bar, H₂/CO=1, GHSV =3.4 Lg⁻¹h⁻¹.

Comment: The catalytic results were obtained by decreasing temperature in the row 350-325-300-275-250°C.

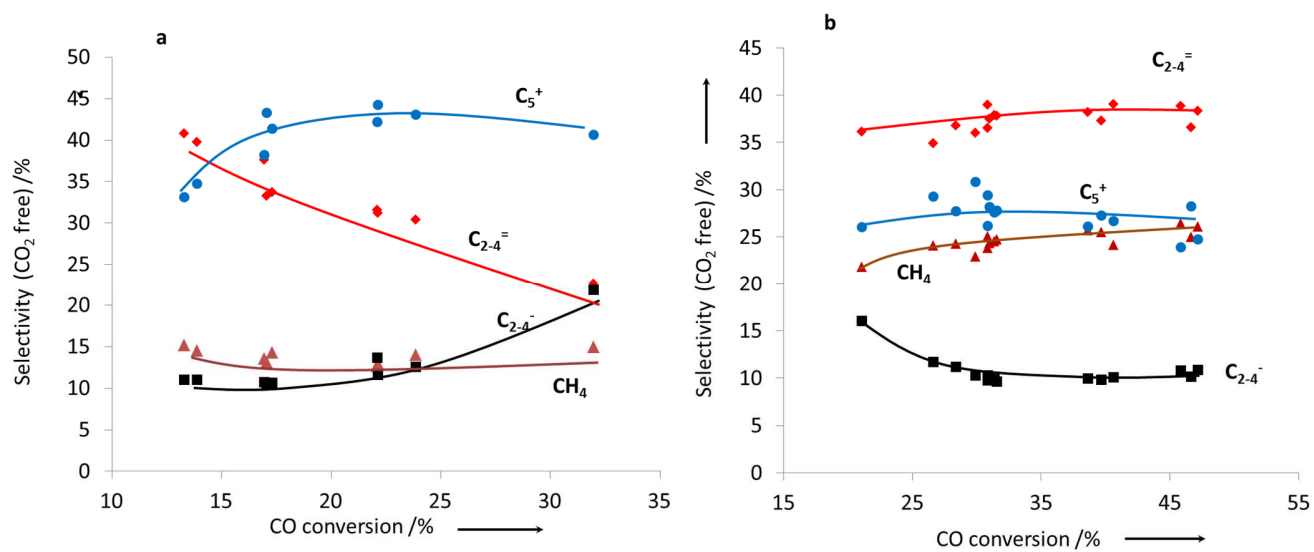


Figure S2. Selectivity to hydrocarbons versus CO conversion over Fe/SiO₂ (a) and FeBi/SiO₂ (b) catalyst. Reaction conditions: P= 10 bar, H₂/CO=1, GHSV =1.5-27 Lg⁻¹h⁻¹, T=350 °C.

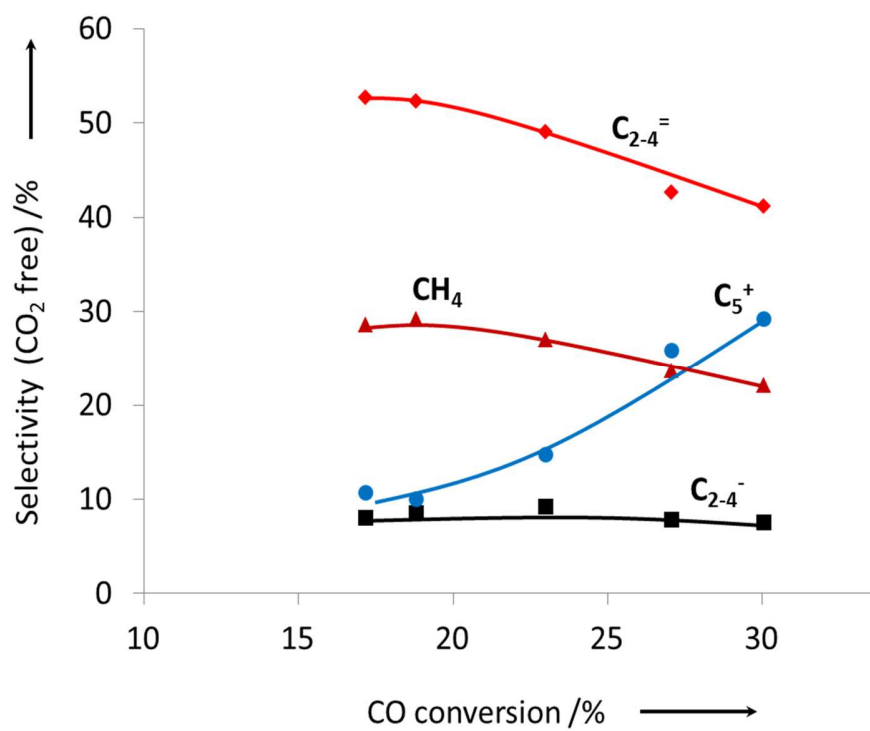


Figure S3. Selectivity to hydrocarbons versus CO conversion over the FeBi/SiO₂ catalyst. Reaction conditions: P= 1 bar, H₂/CO=1, GHSV =1.5-7.5 Lg⁻¹h⁻¹, T=350 °C.

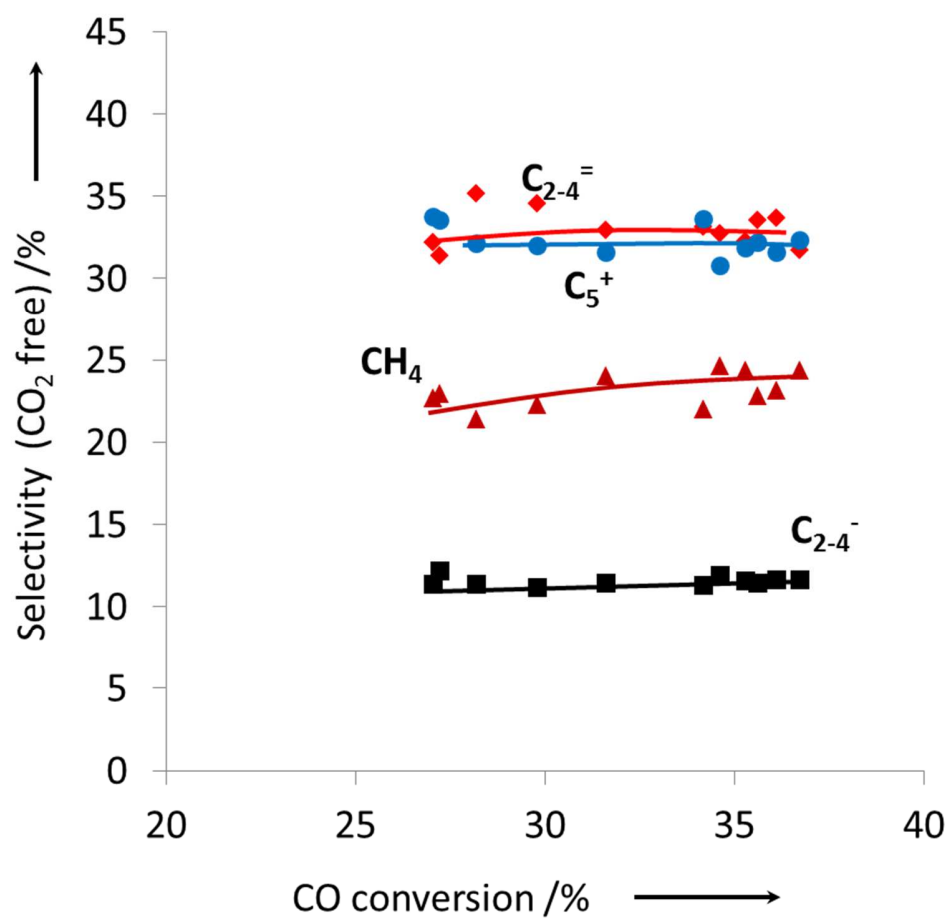


Figure S4. Selectivity to hydrocarbons versus CO conversion over FePb/SiO₂ catalyst. Reaction conditions: P= 10 bar, H₂/CO=1, GHSV =1.5-27 Lg⁻¹h⁻¹, T=350 °C

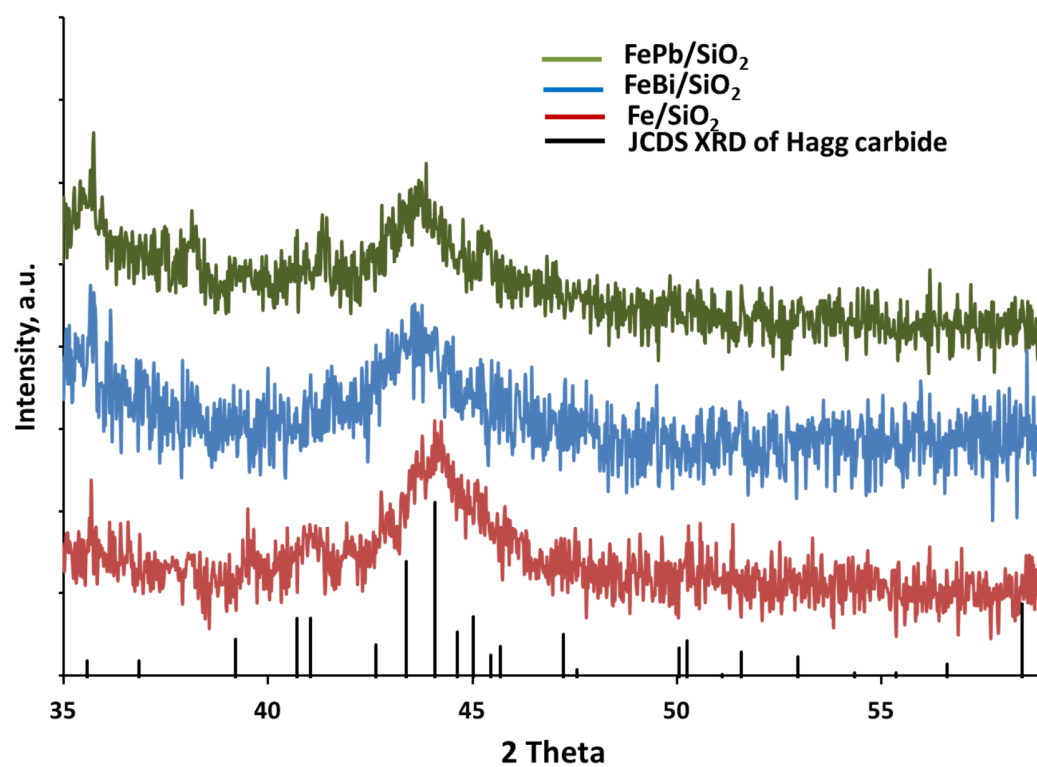


Figure S5. XRD patterns of the catalysts after FT reaction

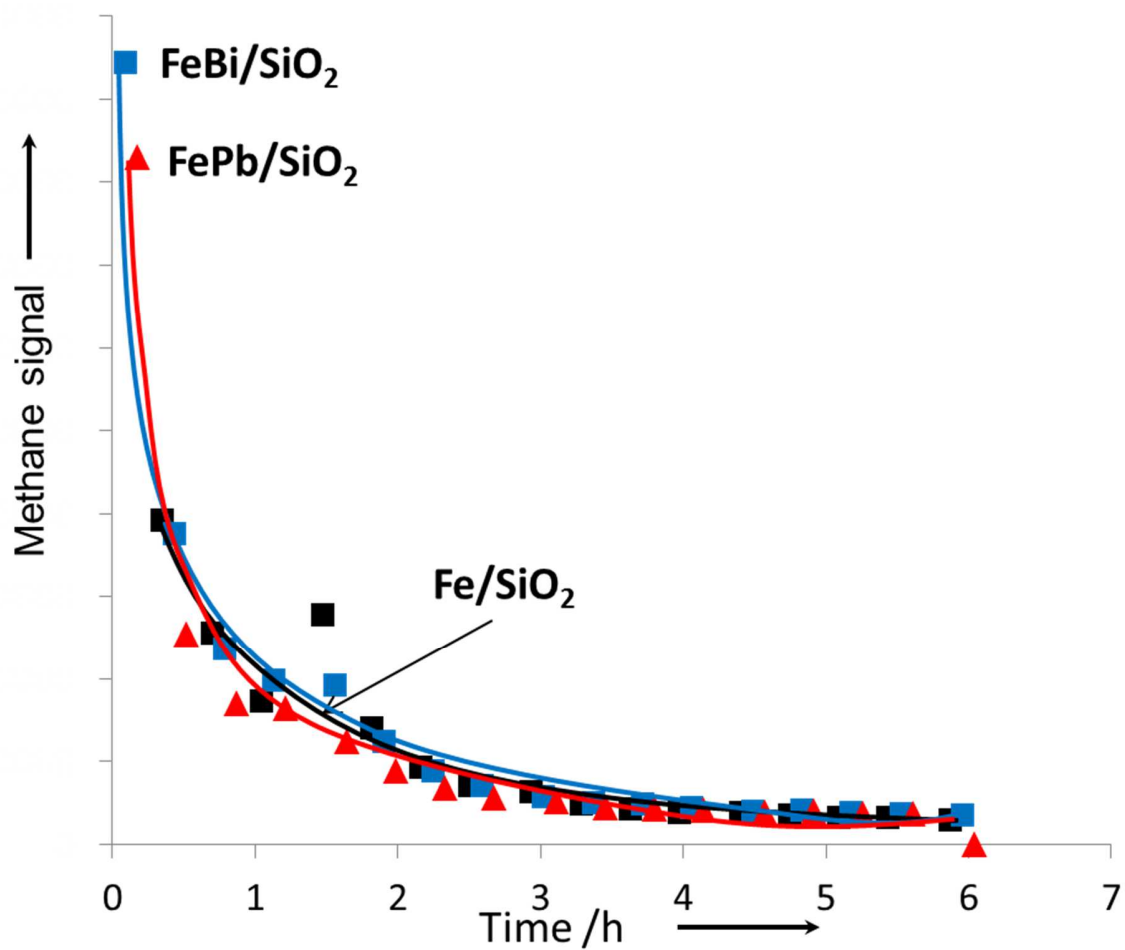


Figure S6. Methane formation rate in static hydrogenation of carbidized Fe/SiO₂, FePb/SiO₂ and FeBi/SiO₂

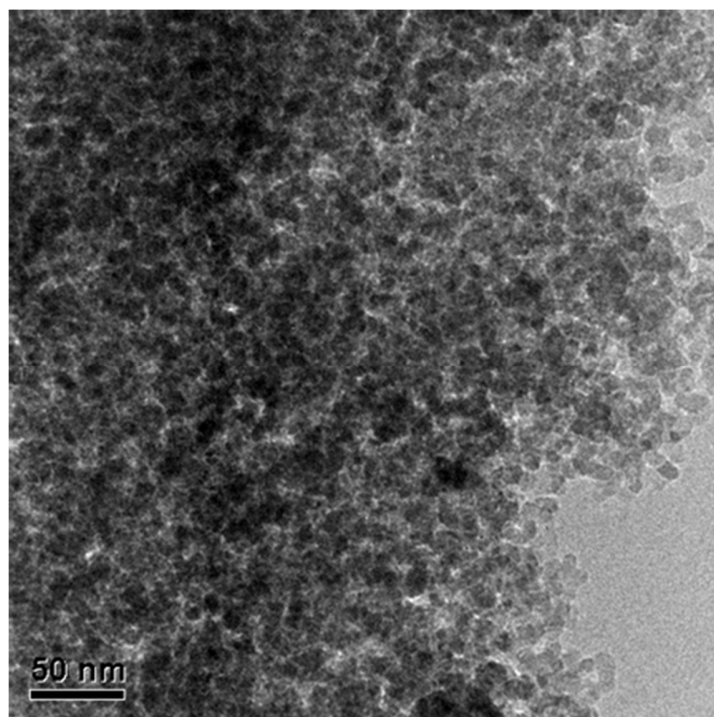


Figure S7. TEM image of Fe/SiO₂

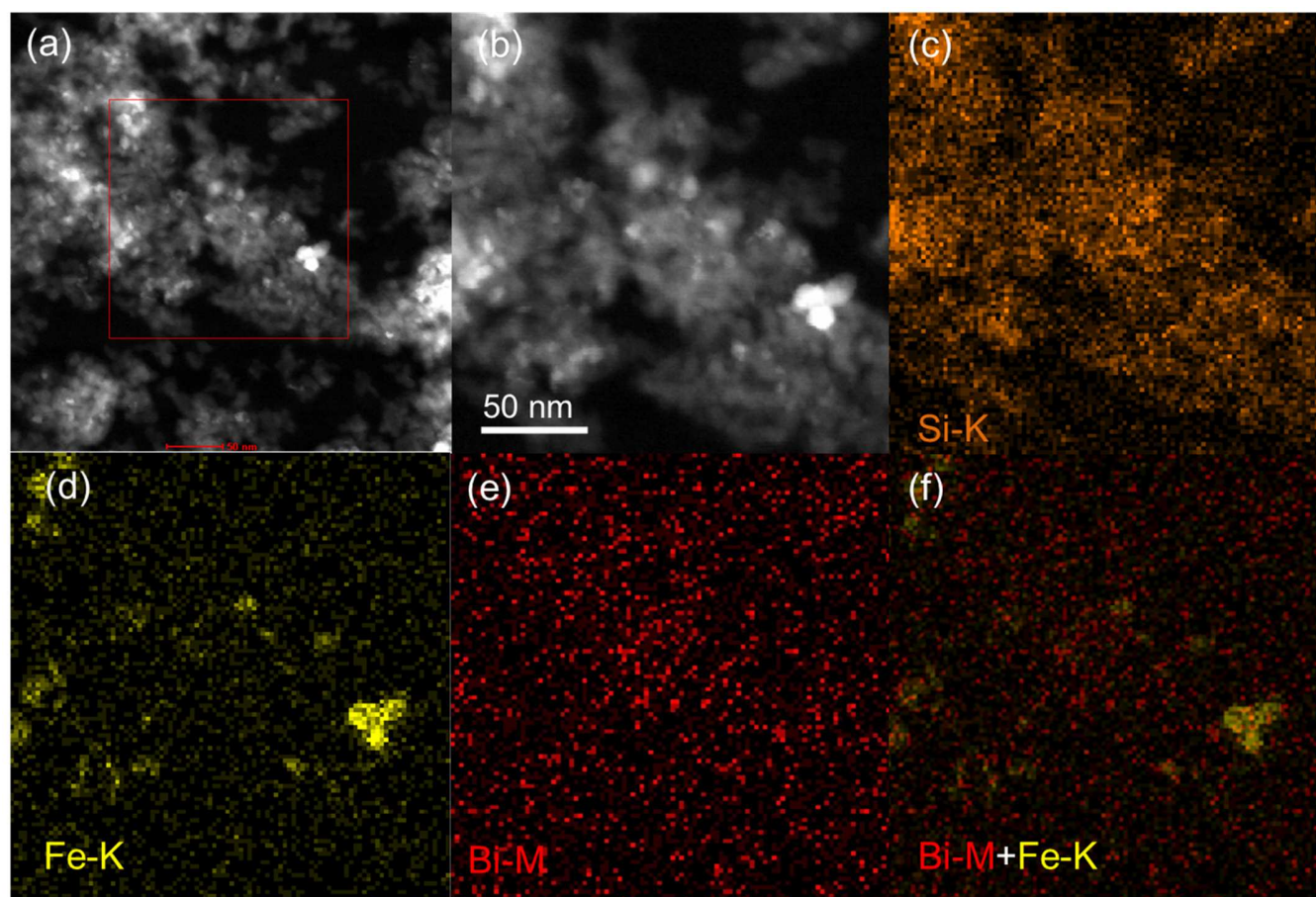


Figure S8. TEM image of Fe/SiO₂ + Bi/SiO₂ (a, b) before reaction with mapping of Si (c), Fe (d), Bi (e) and sum of Bi and Fe (f)

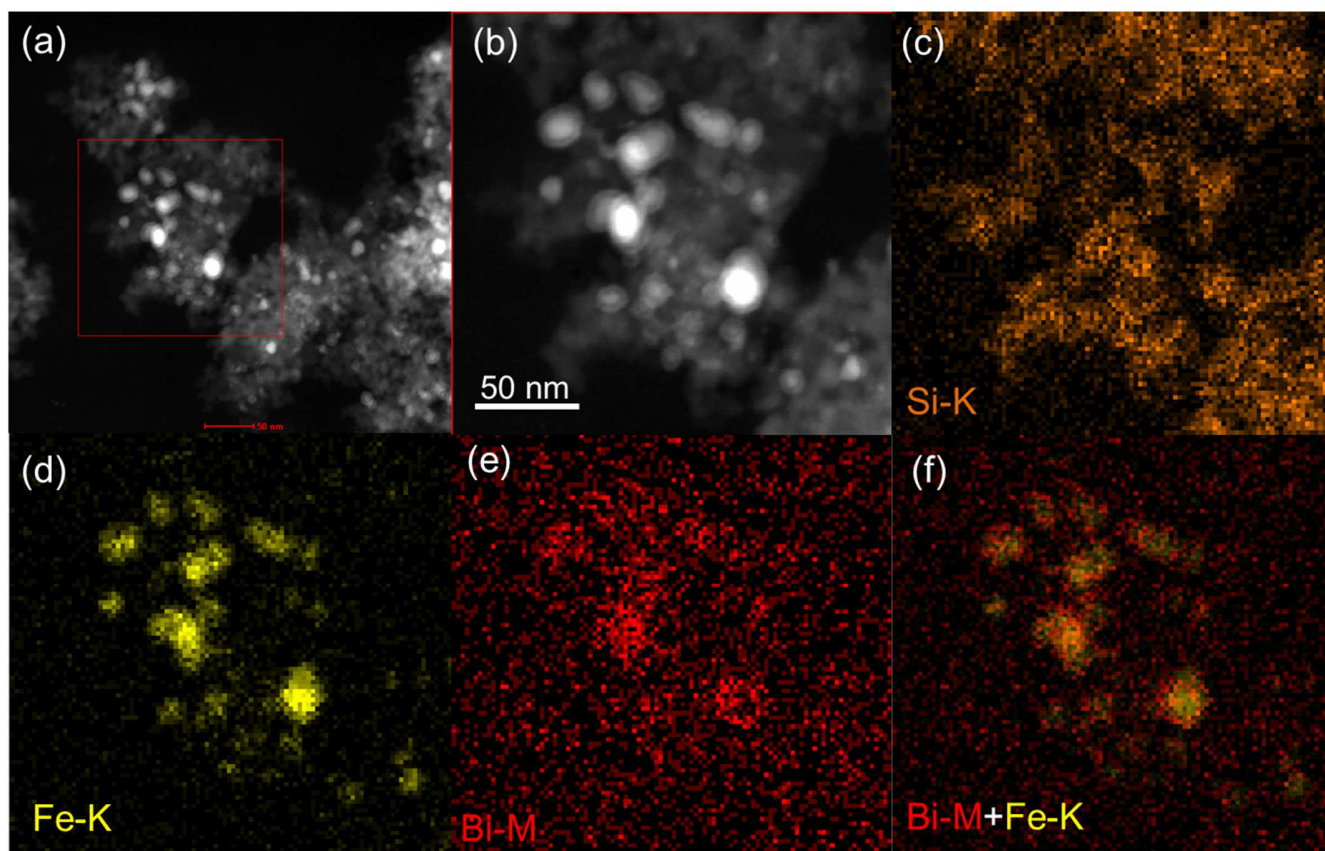


Figure S9. TEM image of Fe/SiO₂ + Bi/SiO₂ (a, b) after reaction with mapping of Si (c), Fe (d), Bi (e) and sum of Bi and Fe (f)

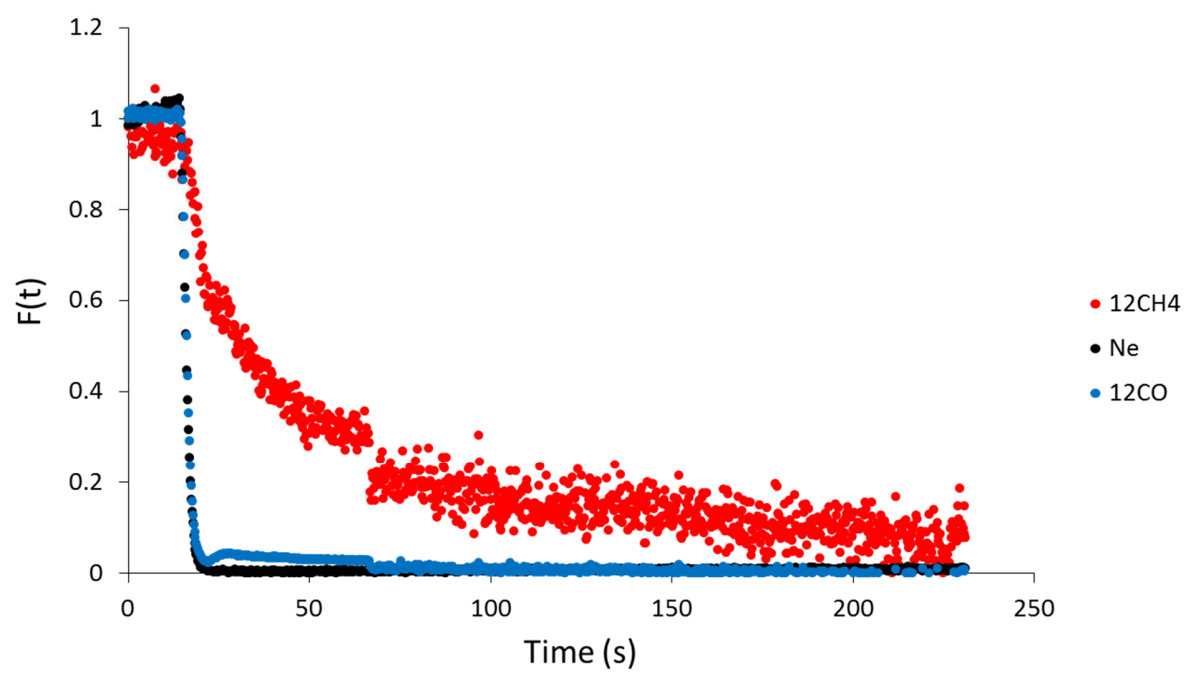


Figure S10. SSITKA results over FeBi/SiO₂. Reaction conditions: P= 1 bar, H₂/CO=2, GHSV =8.4 Lg⁻¹h⁻¹, T=350 °C.

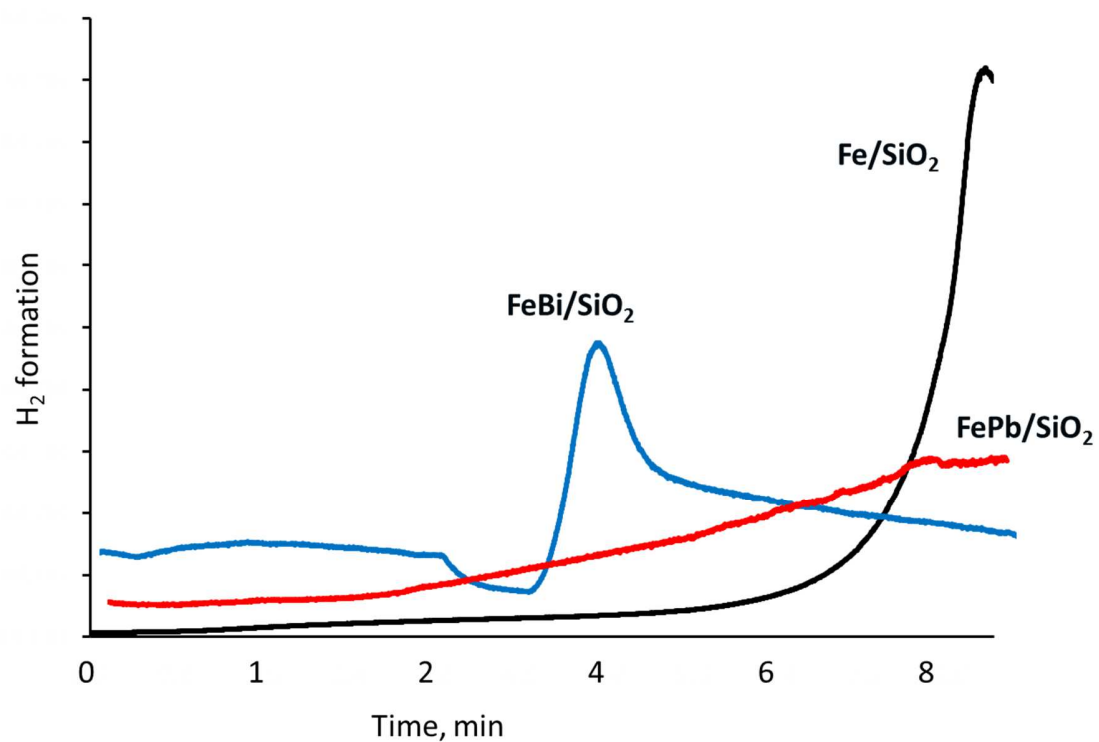


Figure S11. Rate of hydrogen production after exposure of the activated silica supported iron catalysts to water vapours. Reaction conditions: $P = 1$ bar, H_2O flow = 0.3 ml/h, $T = 350$ °C.